**SIADS 696 Milestone II Project Report**

# Fraud Identification in CFPB Complaints Data
# And
# Characteristics of Mortality in Drug Overdose Data in the United States

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## Introduction

### Problems to Solve

This project solves two separate problems that use different datasets. The first problem is how to identify reports of fraud in Consumer Finance Protection Bureau (CFPB) data using supervised learning. The second and entirely separate problem is how to identify patterns in county-level demographic and drug overdose mortality data from the Centers for Disease Control (CDC) Wide-ranging Online Data for Epidemiologic Research (WONDER) and Census American Community Survey (ACS) data using unsupervised learning. The two different groups of data were chosen to better match the learning techniques to the problems and datasets.

### Impact of Solutions

Creating a model that can detect fraud based on the characteristics and narrative description of the consumer complaint allows for more rapid and accurate identification of the nature of the complaint in comparison to a manual review. This facilitates a faster and more informed response from the Consumer Complaint Protection Bureau, as they will have a better understanding of the nature of the complaints that are submitted to them.

Understanding the characteristics of drug overdose deaths from the CDC WONDER and ACS databases allows for better allocation of support funding, staffing resources, staffing development, prevention education, and functional law enforcement to help prevent overdose deaths.

### Motivations

We were motivated to work on these two data groups to get a better understanding of how the data in the CFPB could be used to detect fraud, independent of the consumer's own report of fraud. This would help to contribute to a better allocation of the resources needed to reduce consumer finance fraud and help to make it less prevalent.

For the Drug overdose CSC Wonder and ACS data, we were motivated to better understand the patterns associated with drug overdose, in an effort to be able to provide recommendations for more efficient targeting of resources to help reduce mortality from drug overdoses.

### Supervised and Unsupervised Methods

The supervised learning section of the report compared three models using the data and engineered features from the data to create a predictive model of fraud from the CFPB data. The specific types of models compared were Logistic Regression, Random Forest, and XGBoost.

The unsupervised portion used Principal Component Analysis, K-Means Clustering, Ward's linkage Hierarchical clustering, and Density-based spatial clustering to analyze the data from CDC WONDER and ACS.

### Main Findings

The main finding from the supervised learning of the CFPB fraud data was that the XGBoost model was the best at predicting allegations of fraud, with the XGBoost model having an area under the curve of 0.95. Additionally, the product categories of "...Debt Collection" and "...Credit Reporting…" were the two categories most strongly associated with fraud.

The main findings for the unsupervised learning position were that there was a negative correlation between the median county home price and overdose deaths. Additionally, it was found that 4 clusters best explained the data, and these clusters correlated strongly with geographic and cultural regions of the United States. Finally, it was found that outliers from the main cluster of overdose deaths appear to have three different groups that can be identified through qualitative analysis, possibly identifying underserved communities.

## Related Works

For the supervised learning portion, the most similar project for the CFPB fraud detection portion of the report is the Kaggle project "Automated Complaint Classification using Machine Learning" by Adarsh Singh (1). It uses the same CFPB dataset to classify complaints into 18 product categories (credit card, mortgage, etc.) with 70% accuracy using Random Forest and TF-IDF.

Our project is different in that it predicts allegations fraud vs non-fraud (binary classification with regulatory implications) rather than product categories, and it compares three models and uses ablation testing to see the robustness of the model.

For the unsupervised learning portion, the published paper, "Trends in Drug-Related Mortality in Adults Aged 40–59 in the United States (2018–2023): A Proxy Analysis of Polypharmacy Using CDC WONDER (Wide-Ranging Online Data for Epidemiologic Research)"(2) the papers used CDC Wonder Data to, "For the years 2018-2023, crude mortality rates were evaluated for middle-aged individuals aged 40-59, classified into five age groups. Yearly trends were assessed to evaluate changes over time… [using cross-section analysis] (2)."

It differs from our unsupervised learning project in that it did not merge the data with the ACS data to get a larger overall understanding of the demographics. It also did not use PCA, K-Means clustering, density-based outlier identification, or geographic locations.

Additionally, the CDC WONDER data was also used to publish the paper, "Alcohol-Induced Mortality in the USA: Trends from 1999 to 2020"(3). This study looked at the CDC WONDER data and evaluated it for, "… alcohol-induced overdose mortality in the USA between 1999 and 2020 by age, sex, race/ethnicity, census region, and type of injury (3)."

It differs from our project in that it did not look at county-level data; it looked at the very broad census regions "...Northeast; Midwest; South; West…" (3). It also "…examined the distribution of alcohol-induced mortality by type of injury and regional and sociodemographic characteristics using counts, percentages, and age-adjusted mortality rates(3)." It did not use machine learning, unsupervised learning techniques, and did not include the larger demographic data of the ACS.

## Supervised Learning

## Data Source

The data source is the Consumer Finance Protection Bureau's Consumer complaint dataset. The data can be accessed and manually downloaded here: https://files.consumerfinance.gov/ccdb/complaints.csv.zip

The file contains 162,873 records with consumer narratives from 2020 to the end of January 2026 and is in zipped CSV format. The important variables contained in the file were the date the complaint was received, type of financial product, the narrative of the consumer complaint, and State the complaint was received from.

## Feature Engineering

Fraud labels to create a target for training were derived exclusively from structured fields (Issue and Sub-issue) using keyword matching (e.g., “fraud,” “identity theft,” “unauthorized transaction”). Narrative text was not used to construct the label in order to prevent label leakage. A full list of the variables can be seen in Figure 21. List of features for the supervised learning from the CFPB.

Specifically, three categories of features were constructed:

Text Features, including TF-IDF vectorization of complaint narratives, with unigrams and bigrams for a maximum of 1000 features. Stop-word removal and minimum document frequency filtering were also applied to maximize the eventual usefulness of the model.

Categorical features were also constructed. This included one-hot encoding for the product the complaint was related to, and the state the complaint was reported in.

Finally, numeric features were also created, including the identification of the month, quarter, and day of week. The complaint’s key component is the narrative section, so a length count was added with a word count.

These features were combined into a hybrid feature matrix to capture both linguistic and contextual fraud signals.

Please note that the train/test split was done before cleaning and feature engineering that could cause data leakage were applied.

## Methods

The dataset was sorted chronologically before the 80/20 train-test split was completed. This simulates real-world deployment, where models must predict future complaints based on past data, and prevents data leakage.

Then three diverse supervised models were explored. These models represent fundamentally different learning mechanisms (linear probabilistic, bagging-based tree ensemble, boosting-based ensemble), allowing us to compare the performance relative to each other and select the best one. Specifically, we chose the following three models to examine: Logistic Regression, Random Forest, XGBoost

The Logistic Regression was chosen because it provides an interpretable baseline and establishes a performance floor. Comparatively, the Random Forest was selected to capture nonlinear interactions between narrative text and structured metadata. Additionally, the XGBoost was included as a strong boosting benchmark to test whether sequential error correction improves fraud detection performance. Finally, the hybrid feature representation was chosen because fraud complaints exhibit both linguistic patterns (e.g., “unauthorized charge”) and structured risk indicators (e.g., product category).

These models then had their hyperparameters tuned for sensitivity using 5-fold cross-validation. The tree-based models had variations for their number of trees and maximum depth applied

## Evaluation

Fraud complaints represent approximately 3.8% of the dataset, creating strong class imbalance. Accuracy is therefore not an appropriate primary metric and F1 Score, the harmonic mean of precision and recall, was chosen. Specifically, it was selected because false negatives (missed fraud) are costly, and false positives increase investigation workload. F1 score allows for evaluation of both. Although F1 score is excellent for this application, we also wanted to fully evaluate the model, so we used Precision, Recall, and ROC-AUC to evaluate it as well.

The class imbalance in the data, with far more non-fraud cases than fraud cases made it necessary to set the default threshold for which the classifiers apply labels to something other than 0.5.

In order to discover what the best threshold may be, two alternative threshold strategies were evaluated for each model. The first was the best F1 threshold, which looked at the threshold that maximized F1. For XGBoost, the best threshold was identified as 0.78, resulting in a fraud F1 of 0.51. For the Random Forest model, the best threshold was determined to be 0.65, with a max F1 of 0.45. For the Logistic Regression, the threshold was determined to be 0.86 with an F1 score of 0.38. What this demonstrates is that threshold optimizations can significantly improve fraud detection.

### Overall Results for Supervised Learning (5-Fold Cross-Validation)

Using 5-fold cross-validation on the training data with default thresholds (Table 1):

| Model | Mean F1 | Std F1 |
|---|---|---|
| **Random Forest** | 0.3239 | 0.1070 |
| **Logistic Regression** | 0.2554 | 0.0389 |
| **XGBoost** | 0.0590 | 0.0268 |

Random Forest achieved the highest mean CV F1 (0.3239), though with moderate variance (std ≈ 33% of its mean), suggesting sensitivity to fold composition. Logistic Regression showed the most stable performance with low standard deviation. XGBoost's low CV F1 (0.0590) reflects that, at the default 0.5 threshold, its predicted probabilities are not well-calibrated for the severe class imbalance. However, XGBoost's strong ROC-AUC (0.9472) indicates superior ranking ability. The discrepancy between CV F1 and ROC-AUC suggests that threshold selection — rather than model discrimination capacity — drives the low CV F1. When thresholds are optimized (Section 4), XGBoost achieves the strongest Fraud F1 on the test set.

### Final Model Performance (Test Set)

Using each model's optimized F1 threshold on the held-out test set (Table 2):

| Model | Threshold | Fraud Precision | Fraud Recall | Fraud F1 | ROC-AUC |
|---|---|---|---|---|---|
| **XGBoost** | 0.7788 | 0.51 | 0.50 | 0.51 | 0.947 |
| **Random Forest** | 0.6498 | 0.47 | 0.44 | 0.45 | 0.932 |
| **Logistic Regression** | 0.8618 | 0.36 | 0.40 | 0.38 | 0.924 |

XGBoost achieved the strongest fraud detection performance with the highest Fraud F1 (0.51), the highest precision (0.51), and the highest ROC-AUC (0.947). It detected 569 of 1,127 fraud cases while maintaining balanced precision and recall. Random Forest placed second with Fraud F1 of 0.45 (precision 0.47, recall 0.44). Logistic Regression at its optimized threshold achieved 0.38 Fraud F1, with lower precision (0.36) despite reasonable recall (0.40).

**Key Observation:** XGBoost’s performance advantage may stem from its sequential boosting process, in which each tree focuses on correcting errors from prior trees. This structure can allow the model to capture minority-class patterns that

may not be fully represented through bagging-based ensembles like Random Forest. While this mechanism does not guarantee superior fraud detection, the empirical results suggest that XGBoost is better able to leverage subtle interactions in the sparse feature space.

### Feature Importance and Ablation Analysis

Feature importance was evaluated using Random Forest Gini scores. The most influential structured features included credit reporting, debt collection, and money transfer product categories, along with the temporal feature *Year*, suggesting that certain product types and time trends are associated with higher fraud likelihood.

Key narrative features included TF-IDF terms such as *"fraud," "identity," "bank," "inaccurate,"* and *"reporting,"* indicating that explicit fraud-related language contributes meaningful predictive signal.

Overall, the results suggest that effective fraud detection may depend on both structured complaint metadata and linguistic patterns in the narrative text, rather than either source alone.

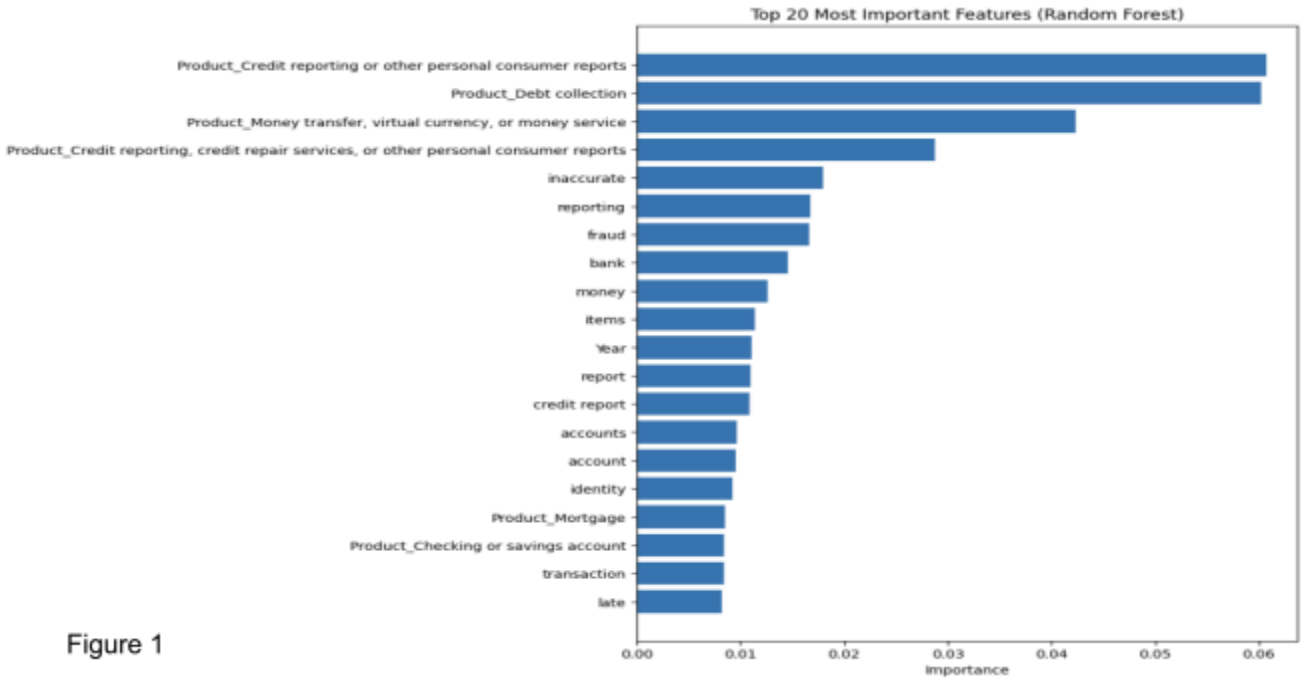



Figure 1

### Ablation Study

To assess the contribution of each feature group, three Random Forest models were trained with optimized thresholds (Table 3):

| Feature Set | Precision | Recall | F1 |
|---|---|---|---|
| **Text Only (TF-IDF)** | 0.28 | 0.35 | 0.31 |
| **Structured Only (Product, State, temporal, meta)** | 0.36 | 0.43 | 0.39 |
| **Hybrid (All features)** | 0.47 | 0.44 | 0.45 |

Neither text nor structured features alone achieved optimal performance. The hybrid model improved F1 from 0.39 (best single-source model) to 0.45, indicating complementary signal between narrative text and structured metadata. Text features capture explicit fraud-related language, while structured features provide contextual risk information. Combining both allows the model to leverage richer predictive information.

### Sensitivity Analysis

Hyperparameter sensitivity was assessed by varying n_estimators ∈ {100, 200, 300} and max_depth ∈ {10, 20, 30} for Random Forest using 3-fold CV on a 20,000-row subsample (Table 4):

| max_depth | n_estimators=100 | n_estimators=200 | n_estimators=300 |
|---|---|---|---|
| **10** | 0.290 ± 0.010 | 0.286 ± 0.006 | 0.288 ± 0.007 |
| **20** | 0.403 ± 0.018 | 0.394 ± 0.021 | 0.404 ± 0.023 |
| **30** | 0.435 ± 0.029 | 0.437 ± 0.027 | 0.444 ± 0.025 |

Increasing n_estimators from 100 to 300 produced less than 1% variation in F1 across all depths.

These results suggest that model expressiveness (tree depth) is more important than ensemble size for capturing interactions in the sparse TF-IDF + structured feature space. Additional trees provide diminishing returns once sufficient ensemble diversity is achieved.

Threshold sensitivity further highlights the precision–recall tradeoff. At its optimized threshold (0.78), XGBoost achieves balanced performance (Precision ≈ 0.51, Recall ≈ 0.50). At the default 0.5 threshold, however, it produces relatively few fraud predictions due to conservative probability calibration under class imbalance, underscoring the importance of threshold tuning.

### Tradeoffs Identified

### Precision vs. Recall Tradeoffs

At its optimized threshold (0.78), XGBoost achieves balanced performance (Precision = 0.51, Recall = 0.50). In contrast, Logistic Regression at its default threshold achieves high recall (0.89) but low precision (0.18), meaning most flagged complaints are not fraud. The preferred operating point depends on whether the priority is fraud coverage (maximizing recall) or investigation efficiency (controlling false positives). XGBoost at 0.78 provides a practical middle ground, capturing roughly half of fraud cases while maintaining manageable false positive rates.

### Feature Representation Tradeoff

Text-only modeling (F1=0.31) misses structural fraud patterns embedded in product and geographic metadata. Structured-only modeling (F1=0.39) misses linguistic nuance — explicit fraud language like "unauthorized" or "stolen identity." The hybrid approach (F1=0.45 for RF, 0.51 for XGBoost) demonstrates that neither signal source is redundant; the improvement over the best single-source model justifies the added feature engineering complexity.

### Model Complexity vs. Interpretability

Logistic Regression offers full coefficient interpretability and achieved the highest default-threshold recall (0.89) at the cost of very low precision (0.18). Random Forest provides moderate performance (Fraud F1=0.45) with interpretable feature importance. XGBoost achieves the best Fraud F1 (0.51) and best ROC-AUC (0.947) but is the least interpretable. For this

fraud detection system, XGBoost was selected because its superior detection performance outweighs the interpretability cost, and feature importance from the Random Forest model can supplement stakeholder explanations.

### CV Stability vs. Test Performance

An important tradeoff exists between cross-validation stability and optimized test performance. Random Forest had the best CV F1 (0.3239) but ranked second on test. XGBoost had the lowest CV F1 (0.0590) at the default threshold, yet achieved the best test-set Fraud F1 (0.51) after threshold optimization. This demonstrates that CV at a fixed threshold can be misleading when models have very different probability calibration characteristics. XGBoost's high ROC-AUC (0.947) confirms strong ranking ability that is realized through proper threshold selection.

### Failure Analysis

Failure analysis was conducted on the Random Forest model at its optimized threshold (0.6498), which produced 635 false negatives (missed fraud) and 564 false positives (incorrectly flagged). The patterns identified also apply to XGBoost, which produced 558 false negatives and 552 false positives at its optimized threshold. Three specific failure examples from each category were examined.

### False Negatives (Missed Fraud)

Three representative cases highlight common failure modes:

- **Legalistic framing (Record #131 — Credit reporting, CA):**
  The complaint uses formal, procedural language (e.g., “absence of concrete evidence”) rather than explicit fraud terms. Because the model relies heavily on vocabulary such as *“fraud,” “unauthorized,”* and *“stolen,”* regulatory-style disputes without these keywords may be underdetected.

- **Formal correspondence tone (Record #151 — Checking/Savings, MI):**
  A structured, letter-style complaint addressed to CFPB resembles a standard dispute rather than a fraud narrative, reducing detectable fraud-specific signals.

- **Anonymization effects (Record #177 — Checking/Savings, NY):**
  Although the complaint clearly describes hacking and unauthorized access, anonymized tokens (e.g., “x x year”) may weaken lexical signal strength, contributing to misclassification.

These cases suggest that the model may underperform when fraud is described indirectly, formally, or with masked numeric details.

### False Positives (Incorrectly Flagged)

Three examples illustrate overprediction patterns:

- **Account freeze language (Record #12 — Debt collection, TX):**
  Terms like “frozen account” overlap with fraud-related vocabulary but refer to a routine dispute.

- **Unexpected debit card (Record #82 — Checking/Savings, GA):**
  Language resembling identity theft scenarios may trigger fraud classification, even when the complaint reflects normal servicing issues.

- **Held funds in money transfer (Record #192 — MA):**
  Product categories with high fraud prior probability (e.g., money transfer) combined with ambiguous terms like "held" can increase false positives, even when the issue concerns access disputes rather than fraud.

Overall, false positives often arise from vocabulary overlap between fraud and non-fraud complaints, particularly in high-risk product categories.

### Patterns Across Failures

Average narrative length for false negatives (911 characters) was similar to correctly detected fraud (894 characters), suggesting that length alone is not a meaningful discriminator. False positives were slightly shorter on average (752 characters), indicating that shorter complaints with ambiguous vocabulary may lack sufficient context for accurate classification.

### Potential Improvements

Several modeling refinements could address observed failure modes:

- **Contextual embeddings (e.g., BERT or sentence transformers)** may better distinguish legalistic fraud complaints from routine billing disputes by capturing semantic relationships beyond TF-IDF.

- **Phrase-level modeling** could identify multi-word fraud indicators (e.g., "without my permission," "opened in my name") that bag-of-words representations may dilute.

- **Product-specific classifiers** may account for vocabulary differences across complaint categories, reducing ambiguity in high-risk product types.

### Final Model Selection

XGBoost was selected as the final modeling framework because it achieved the strongest fraud detection performance across evaluation stages. At its optimized threshold, XGBoost reached a Fraud F1 of 0.51 and ROC-AUC of 0.947, with balanced precision and recall (0.51 / 0.50), outperforming Random Forest (F1 = 0.45) and Logistic Regression (F1 = 0.38).

After applying SMOTE to address class imbalance (~3.8% fraud rate), performance improved further. XGBoost + SMOTE achieved the highest overall Fraud F1 (0.54) and ROC-AUC (0.950), increasing recall to 75% at the cost of reduced precision. This configuration captures substantially more fraud cases and is preferable when minimizing missed fraud is the priority.

Although Random Forest provided useful feature importance insights and strong cross-validation stability, it did not match XGBoost's ability to model nonlinear interactions in the high-dimensional feature space. Therefore, XGBoost + SMOTE is selected as the final model, with the non-SMOTE configuration retained when precision must be emphasized.

#### Supplementary: SMOTE Oversampling Experiment

With a ~3.8% fraud rate, the dataset is highly imbalanced. To evaluate whether minority oversampling improves detection, SMOTE was applied to the training set only, expanding it to a 50/50 class balance (254,734 rows). The test set remained unchanged.

**SMOTE Comparison Results** (Table 5)

| Model | Threshold | Fraud P | Fraud R | Fraud F1 | AUC |
|---|---|---|---|---|---|
| RF (Original) | 0.6498 | 0.4659 | 0.4366 | 0.4508 | 0.9323 |
| RF + SMOTE | 0.4741 | 0.4586 | 0.5306 | 0.4920 | 0.9459 |
| XGB (Original) | 0.7788 | 0.5076 | 0.5049 | 0.5062 | 0.9472 |
| XGB + SMOTE | 0.1091 | 0.4252 | 0.7542 | 0.5438 | 0.9503 |

**Key Findings:** SMOTE improved Fraud F1 for both models (RF: 0.45 → 0.49; XGBoost: 0.51 → 0.54). The largest impact was on recall: XGB + SMOTE detected 75.4% of fraud cases (vs. 50.5% without SMOTE), though precision declined from 0.51 to 0.43. The optimized threshold shifted substantially (0.78 → 0.11), reflecting probability redistribution under rebalancing.

AUC improved modestly (0.9472 → 0.9503), suggesting that SMOTE slightly enhanced ranking performance, but its primary effect was shifting the operating point toward higher recall.

**Practical Implications**

Model selection depends on operational priorities. If minimizing missed fraud is paramount and flagged cases undergo manual review, XGB + SMOTE may be preferable. If false positives are costly in automated workflows, the original XGBoost model offers a more precision-balanced alternative.

## Unsupervised Learning

## Data Sources

An Application Program Interface (API) was used to download data from the Centers for Disease Control (CDC) from the Wide-ranging Online Data for Epidemiologic Research (WONDER) and the United States Census Bureau for the American Community Survey (ACS). The CDC WONDER data is available for download via the following:

https://wonder.cdc.gov/mcd-icd10-expanded.html

The CDC WONDER data has substantial restrictions on the number of files that can be accessed at once. This project attempted to use the API, but found that it was difficult to use consistently. As a result, manual downloading is possible and recommended. The data downloaded returns in a CSV format and two separate groups of data were downloaded. The first set of data looked at all deaths in the county and included the important variable of the population of the county. The second set of data was data by race, and set to identify the underlying cause of death as being the UCD Drug/Alcohol Induced Causes, and then the population specific to Drug-induced cause was used (ICD-10 codes: Drug overdose, specifically X40-X44, X60-X64, X85, Y10-Y14). This dataset allowed us to identify the race of the population who had suffered a death from overdose. Both of these datasets also came with location identification through the FIPS codes, state names, and county names.

The data covered 2018 to 2023. For the file that covered all deaths, 9,585 records were retrieved. For the files that covered the deaths by race, 11,567 records were accessed.

The Census data can be accessed via:

https://api.census.gov/data

An API retrieved multiple reports that can be seen in the appendix in figure 22. List of specific reports from the Census ACS database. The returned format of the API was JSON, and about 16,104 records were returned for the time range of 2018 to 2022. The important features were Median Household Income, Median Home Value, Median Age, Poverty Rate, Unemployment Rate, and Multiple columns for level of college completed.

## Feature Engineering

The CDC data was cleaned for column formatting, and data entries with fewer than 20 deaths in the entry were removed due to the CDC's historical practice of noting them as unreliable (4). This allowed us to have data for the population for the counties by year. Data based on the race of the population was then downloaded to allow for the death count by race to be identified. The two datasets were then merged on Federal Information Processing Standards (FIPS) codes and the year to create a single dataset that included both deaths by race for the counties and the county population. This then allowed us to calculate a crude death rate based on the number of deaths by race and the county's population.

Demographic data regarding elements like median household income and unemployment rates for the county were taken from the ACS data and matched to the CDC WONDER data with the FIPS code and year. This data was then merged with the CDC WONDER Data to create a single dataset.

The features were selected based on availability in the datasets and expected demonstration of effect due to economic importance or size of the population proportion. Specifically, the CDC Wonder data has limited data that includes the details of location and time (state, county FIPS codes, year) and number of deaths by the ICD-10 code. The ACS data is broader, and for analysis, we chose features that described large portions of the population (race/ethnicities with large proportions of the population in most counties and education level) and were descriptive of key economic factors present in the county (median income, median home value, poverty rate). Average Crude death rate was chosen because it reflected death data that had county/years with less than 20 deaths removed and normalized the data over the years it was available for, thereby ensuring that the clusters would be based on more substantial characteristics of the data, rather than artifact from counties with small populations or outlier events.

The merged dataset then had KNNImputer used based on the 5 nearest neighbors and StandardScaler to prepare it for dimensionality reduction using SciKit learn's Principal Component analysis PCA.

The components from the PCA may also be considered features, as they were used in the K-Means clustering and DBSCAN. The PCA identified 10 principle components in total labelled PC1 through PC10.

The complete list of all final features prior to analysis with PCA can be seen in the appendix in figure 23. List of features in the Unsupervised learning dataset.

### Methods

The first method chosen was Principal Component Analysis. SciKit Learn's PCA was used to complete this analysis. The PCA results were visualized in a PCA Scree plot to show the proportions of how much each component explains the variance. They were also visualized in a PCA biplot to show how the variables related to each other. Specifically, we used PCA to reduce the dimensionality of the data so it could be more easily processed and to understand how the variables related to each other through the PCA Biplot with the components that explained the most variance.

In addition to the PCA analysis, SciKit Learn's K-Means was used to cluster the data. To tune the hyperparameters, K values 2 through 10 were evaluated for the most effective clustering using SciKit Learn's inertias for the Elbow Method, sil_scores for the Silhouette Score, and db_scores for the Davies-Bouldin Index for the values. K-Means clustering was used to identify the clusters of similar characteristics that occurred in the data. We also looked at the features in the clusters after K-Means clustering, to understand the characteristics of the clusters.

As part of the clustering evaluation, hierarchical clustering was also evaluated using SciKit Learn's linkage with the "Ward" method in order to evaluate if K-Means or hierarchical clustering produced better clustering. The dendrogram was visualized and the Rand Index was used to compare the K-means and Hierarchical clustering results. This was done to determine if hierarchical or non-hierarchical clustering would create more effective clusters.

SciKit Learns DBSCAN was used to identify the outliers using the rule of nearest neighbors (K) of 2 times the dimensions of the data. The elbow was identified using the 95th percentile distance for clustering, which allowed us to cluster the majority of the data, and only identify outliers.

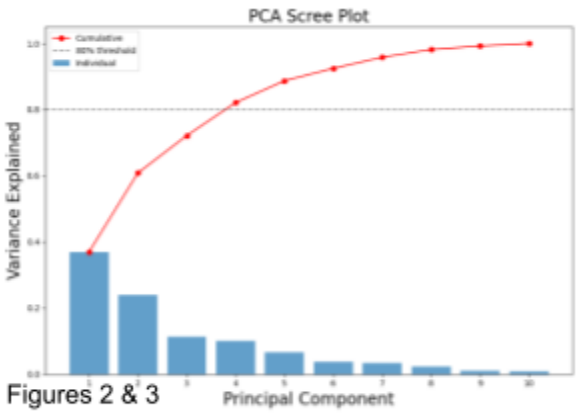
Finally, ablation testing was used to determine the sensitivity of the clustering to the elimination of each variable.

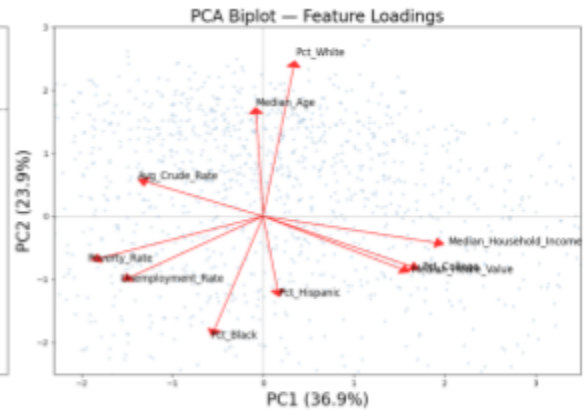
In addition to all the steps above, the combined datasets were then visualized on two different geographic maps of the United States. This allows us to have a practical understanding of what the outcome of the clustering means in relation to the geography of the United States, to contextualize the cluster and outlier data to better understand them from a qualitative perspective, and to better communicate the results to the report's audience.

## Unsupervised Evaluation

### Evaluation Metrics

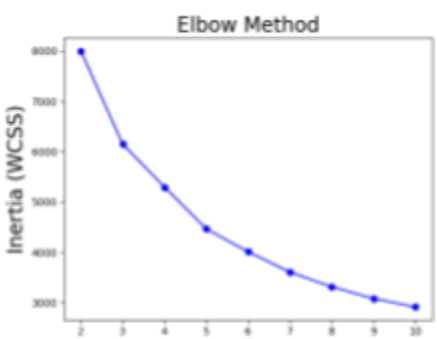
For the PCA, the evaluation metric used was a Scree Plot that looked at how much of the variation in the data was explained by each component. The plot organized the components by most variance to least variance and showed the additive value of how much of the variance each component explained. This was chosen because it allows for a visualization of how much of the variation in the data is explained by the new components, shows how the different components can be combined to reach the basic threshold of 80%, and is a standard way of evaluating the results of PCA.

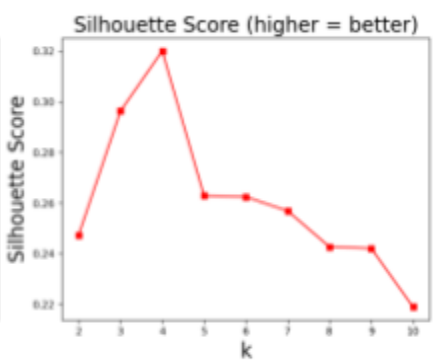



Figures 2 & 3

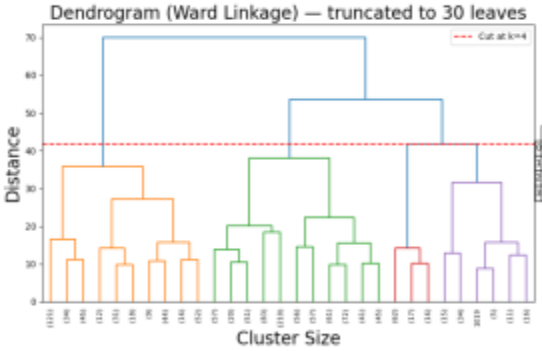
For K-Means, clusters were calculated for K-values 2 through 10. To evaluate the results of these different hyperparameters, we used three different evaluation methods. The first method used was the Elbow method, which allows for a rough understanding of what the value of K should be when looking at what the optimum number of clusters is for use with K-Means. Although the Elbow method allows for a rough understanding of what value for K should be used, it is less precise and less useful for comparison than the Silhouette Score and the Davies-Bouldin score. Both of these scores were used to evaluate the clustering as they look at different aspects of clustering. The Silhouette Score looks at how similar the point is to other points in the cluster it is assigned to, and the Davies-Bouldin looks at how similar the cluster is to the next most similar cluster. The combination of these scores allowed for an understanding of what the optimum values for K should be.

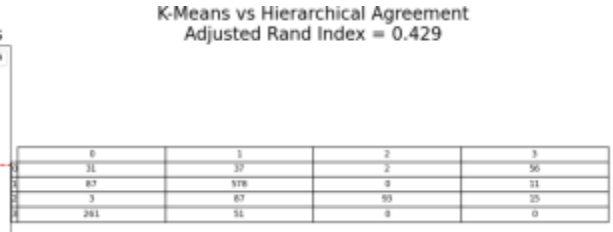



Figures 4 & 5

Additionally, we looked at how the K-Means clustering compared to Ward Linked hierarchical clustering using the Rand index to determine how the different clustering models grouped the same points and the resulting quantitative measure for it. This allowed a measurable result to determine if K-Means or hierarchical clustering were the best method to produce clusters.



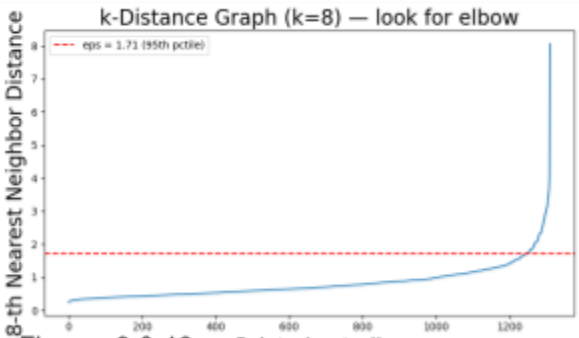

| | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| 0 | 31 | 37 | 2 | 56 |
| 1 | 87 | 578 | 0 | 11 |
| 2 | 3 | 87 | 93 | 15 |
| 3 | 261 | 51 | 0 | 0 |

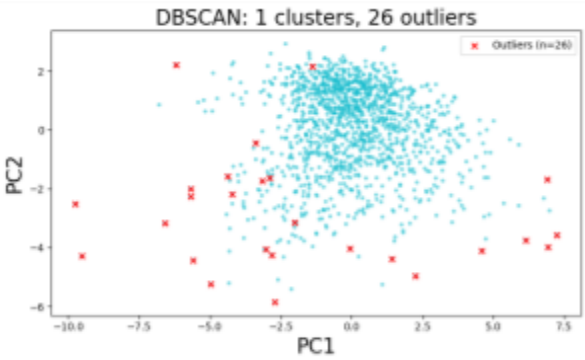
Figures 6 & 7

PCA was selected to understand how the variables available in the data correlate with each other, and to what extent the data can be explained with two of the features created in the analysis. In this case, the maximum amount of the variance that can be explained with two features is 60.8%, making the use of PCA for visual clustering on two dimensions ineffective, and indicating the K-Means clustering on all the components would yield results with more information from the original data. We also used cluster analysis and a heatmap of z-scores in the clusters to understand the characteristics of the clusters.

Figure 8

For DBSCAN, we again used the elbow method, but were already aware of the needed number of neighbors based on the dimensions of the data. We set the eps value to 95%, so we could identify the 5% of the points who had nearest neighbors beyond the threshold. This allowed for a confident identification of the primary cluster and its outliers.




Figures 9 & 10

Finally, the geographic maps allowed for a qualitative evaluation of the data. The ability to identify themes in the data that correlated with expected existing structure that could be reflected in the data confirmed that the thresholds and methods of the analysis provided potentially meaningful insight into the data. The map can be seen in the Discussion section below in Figure 12.

### Summary of Results from the Best Models

For the PCA analysis, the model that explained at least 80% of the variance using the fewest components, did so using the 4 largest components. Specifically, the 4 largest of the 10 components identified explained 82% of the variance in the data. The Biplot of the PCA data showed that there was strong correlation between the following variables:

- White and Median Age
- Median Household Income, Median Home Value, and Percent College

- Unemployment and Poverty Rate

Additionally, there was no meaningful correlation between:

- Average Crude Death Rate and Percent White or Percent Black
- Unemployment and Median Home Value or Percent College

Finally, there was an inverse correlation between:

- Average Crude Death Rate and Median household income, Percent College, or Median Home Value
- Hispanic Ethnicity and Median Age

Please see figures 2 and 3 Scree chart and PCA Biplot for further details.

For the K-Means Clustering analysis, the Silhouette Score provided the sharpest feedback as to the optimum k-value for clustering. After looking at k values 2 through 10, you can see how clearly the k-value of 4 stands out in the graph with a Silhouette Score of 0.32. Please see figure 5, K-Means Silhouette Score Graph for further details. Additionally, the 4 clusters had distinct characteristics that are further described in Figure 8, Cluster z-Score chart.

For the evaluation between K-Means clustering and Ward's Linkage Hierarchical clustering, the adjusted Rand index was 0.43. The details of this analysis can be seen in Figures 6 & 7, Hierarchical Clustering Dendrogram and Cluster comparison chart.

The DBcsan elbow score with 8 nearest neighbors and an eps value at the 95th percentile produces a sharp elbow graph, with only 2% of the counties resulting in outliers. Figures 9 & 10, Elbow graph for DBSCAN eps value at 95th percentile and DBSCAN cluster and outliers.

Finally, for the geographic mapping, K-means clusters reflected metropolitan areas, rural areas, and what has historically been considered to be The South. The outliers from DBSCAN show patterns including metropolitan regions on the East and West Coasts, unique outliers running a line north to south down the center of the country. The details of this map can be seen in the discussion section below or in Figure 12. Geographic map of drug overdose mortality and demographic data with DBSCAN outliers in the appendix and in the Discussion section below.

## Sensitivity Analysis

In addition to the use of Silhouette Score to determine the optimum number for K as noted above in the Methods and Evaluation Metrics sections, we also examined how sensitive the K-Means clustering was to feature choice. To do this, we looked at what the results were if we dropped a feature and continued to use a value of 4 for K with PCA data clustered with K-Means. We evaluated the outcomes by using the Adjusted Rand Index comparing the ablated clusters with the original cluster that contained all the variables. What we found was that the Unemployment Rate was the least important variable, while features like Percent Hispanic and Median Age were more important to clustering. See Figure 11 for further details.

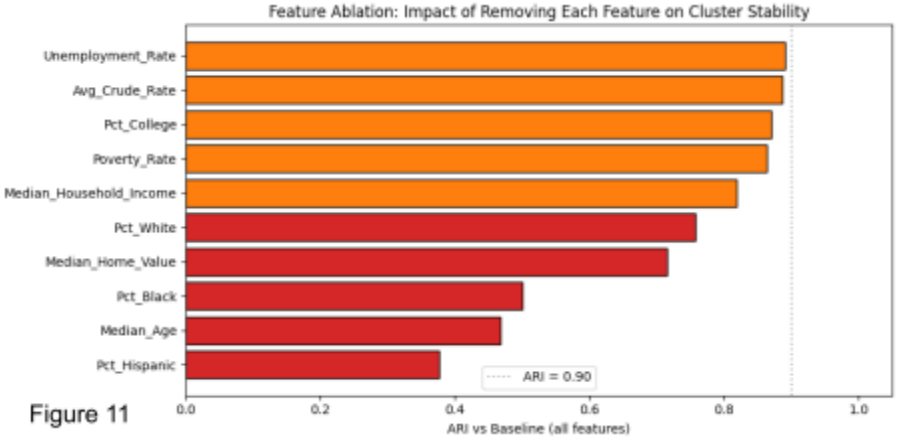



Figure 11

## Discussion

**What was learned from Supervised Learning**

The most important result from the supervised learning portion was how strongly performance depended on threshold calibration and class imbalance handling. Although Random Forest initially appeared strongest under default cross-validation settings, XGBoost ultimately achieved the best fraud detection performance once thresholds were optimized. This highlighted how misleading default settings can be in highly imbalanced datasets.

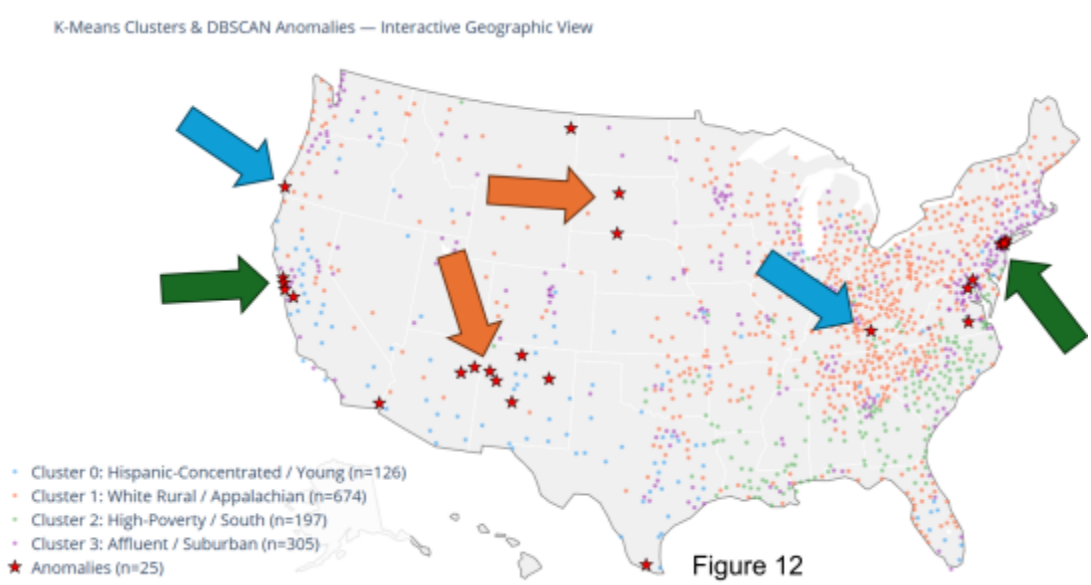
Additionally, the hybrid feature representation clearly outperformed both text-only and structured-only models. Fraud detection depended on both contextual metadata (product category, time trends) and narrative language patterns. The ablation study confirmed that neither feature group alone was sufficient.

Finally, the SMOTE experiment demonstrated that recall could be substantially increased with only modest changes in ROC-AUC. This showed that operating point selection — not just model architecture — plays a critical role in fraud detection performance.

**What was learned from Unsupervised Learning**

The most surprising result was the clear clustering that showed regionalization from the K-Means clustering. When mapped, the K-Means clusters correlated with existing geographic and culturally identified areas of the United States. Urban areas and the section of the United States historically labeled as The South are clearly identifiable on the geographic map. Additionally, the outliers identified on DBSCAN showed further patterning when the mouse over was used to investigate the qualitative elements of the groupings. Outliers with similar demographics are shown with similar colored arrows. The most interesting grouping being those indicated with the orange arrows, which potentially represent a less well-known group of overdose victims. The outliers in those regions are in counties with higher proportions of Native Americans than in other areas of the country. The blue arrows represent a similar grouping of rural areas with little agriculture and notable poverty. Finally, the green arrows are clusters of metropolitan areas with dense populations.

Figure 12. Geographic map of drug overdose mortality and demographic data with DBSCAN outliers.




The challenges that were encountered were the result of the need to evaluate the clustering across multiple criteria. The outcomes of the clustering had to result in meaningful results, and although the Silhouette Scores result in mathematically meaningful clusters, the geographic mapping is fundamentally what allowed for the confirmation that the unsupervised learning produced meaningful results.

With more time and resources, the solution could be extended to look at other diagnoses that are available in the CDC WONDER data. We could easily use this same method to look for patterns in stroke or heart attack diagnoses, or diabetes diagnoses.

## Ethical Considerations

For Supervised Learning CFPB data, the consumer complaints from the CFPB data that were used to train our model contain allegations of fraud and not definitive instances of proven fraud. Care must be taken to ensure that users of the model to predict fraud understand that it is able to identify allegations of fraud, not prove that fraud is the cause of the consumer's report. We would address this by educating the users of the model as to its limitations and intended use. Additionally, this model will have drift over time, as new financial products are developed and new types of fraud are identified on the market. Anyone who makes use of this model will have to ensure the ongoing accuracy of the model.

For the Unsupervised Drug Mortality data section, the data used mortality and demographic data that could be considered sensitive and could potentially be used to identify victims of overdoses in counties with small populations. This was addressed with a data inclusion cutoff of at least 20 deaths in a county in a year. To add to the anonymization, the merging of data over years further prevents meaningful identification of individual victims on the geographic map.

Additionally, caution must be taken to ensure that those who are reading the results understand the clustering to be based on correlation and not an indication of the cause of drug overdose deaths. Specifically, although we are identifying outlier groups with certain regions of the country, counties, and races. The deaths still represent a small number of deaths in comparison to the total numbers of deaths in the county for all races. Inference should not be made about the general well-being of the region or its population based on this report. Additionally, the descriptive demographics are not the cause of drug overdose deaths but are correlated with the deaths and descriptive of possible correlations in the data. The authors of this report clearly state that no inference can be made based on it as to the cause of drug overdose deaths in a region, county, or race/ethnicity.

## Statement of Work

Keerthi Vallabhaneni: Unsupervised and Supervised Learning sections coding and report writing
Oluwabusayo Okusaga: Supervised Learning section coding and report writing
Scott Dye: Unsupervised Learning section coding and report writing

## References


1. https://www.kaggle.com/code/adarshsng/consumer-complaint-classification-nlp
2. https://pubmed.ncbi.nlm.nih.gov/41333966/
3. https://pmc.ncbi.nlm.nih.gov/articles/PMC10243241/
4. https://wonder.cdc.gov/wonder/help/mcd-provisional.html


**AI Disclosure**

Claude and ChatGPT were used to assist in writing the code for this project

## Appendix

Figure 1. Most Important Features for supervised learning

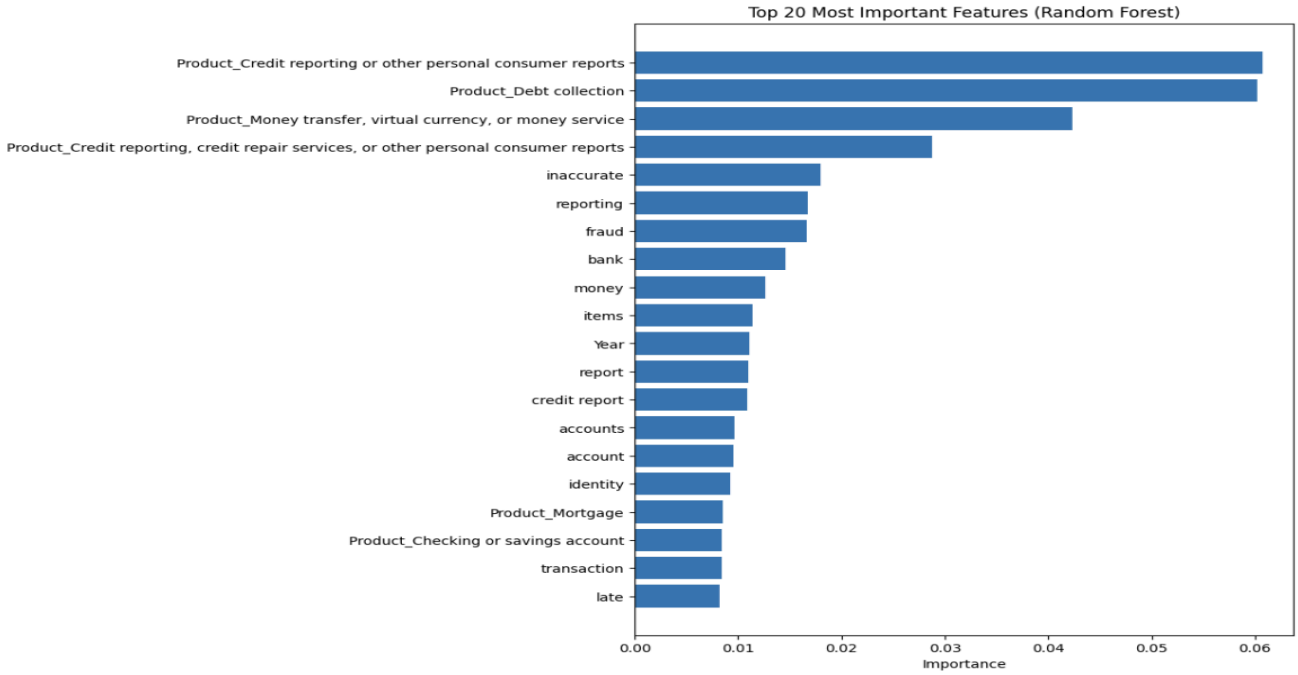



Figure 2. Scree Chart

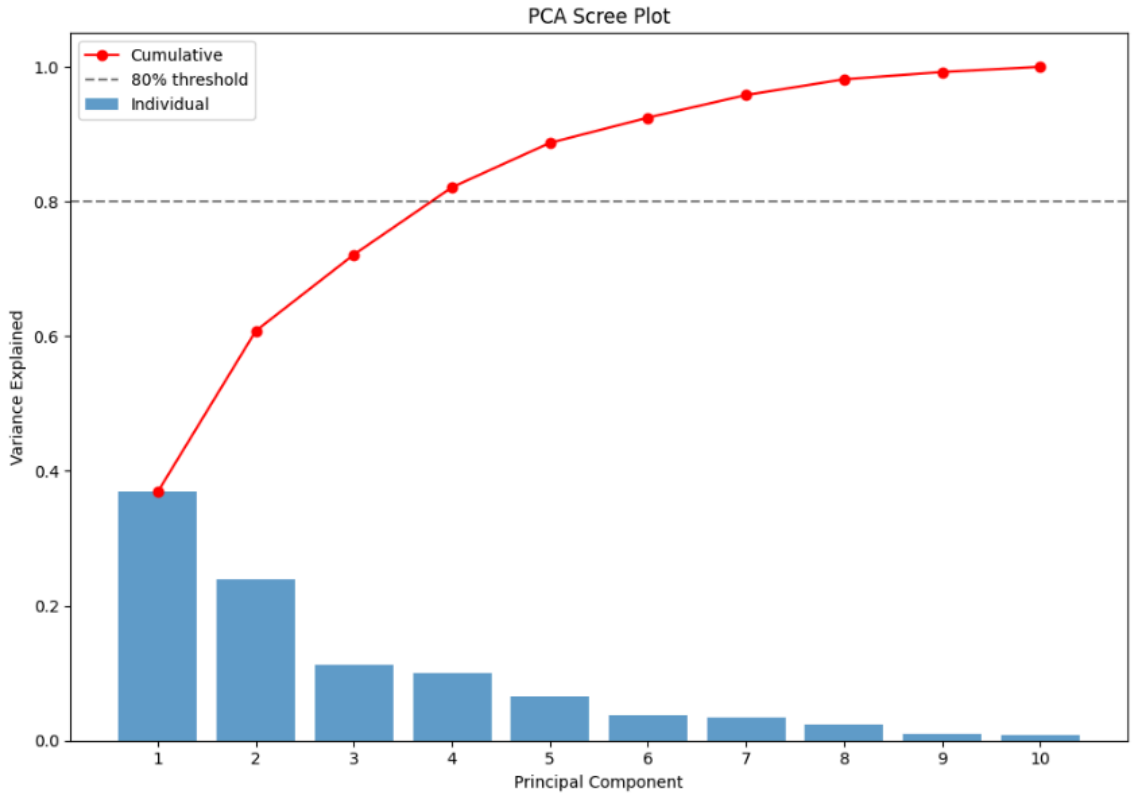



Figure 3. PCA Biplot of PC1 and PC2

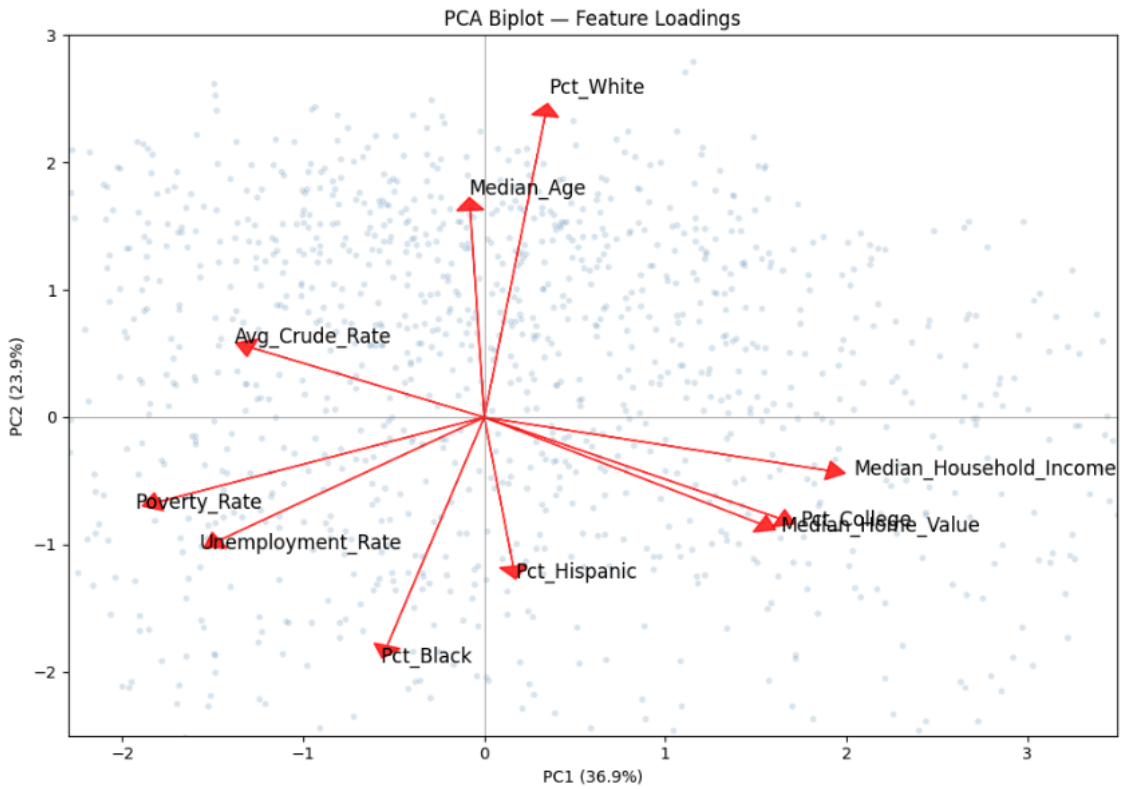



Figure 4. K-Means Elbow Method Graph

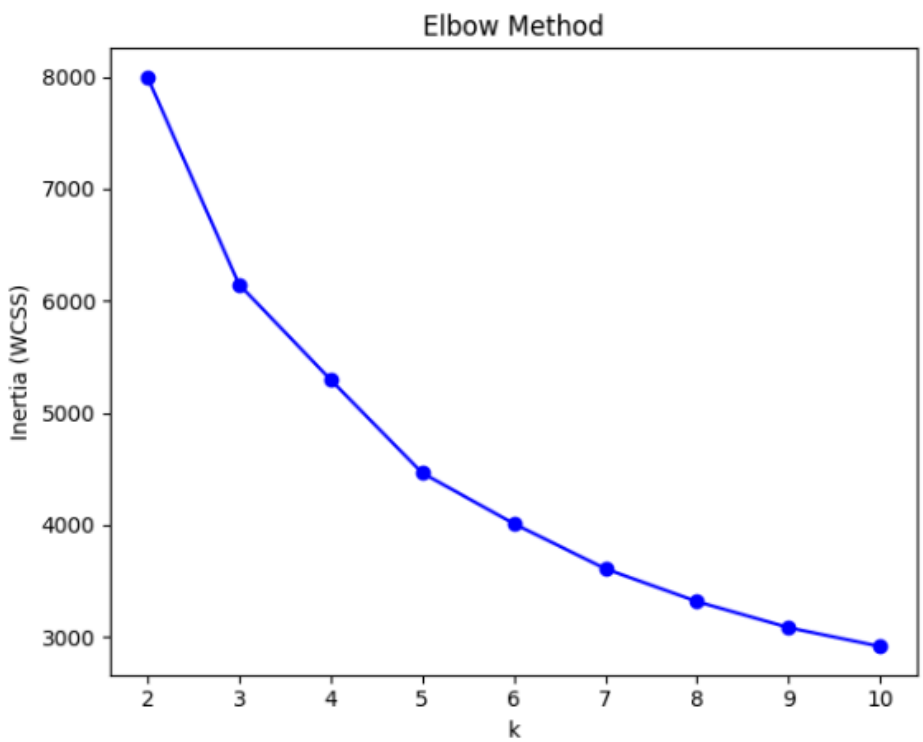



Figure 5. K-Means Silhouette Score Graph

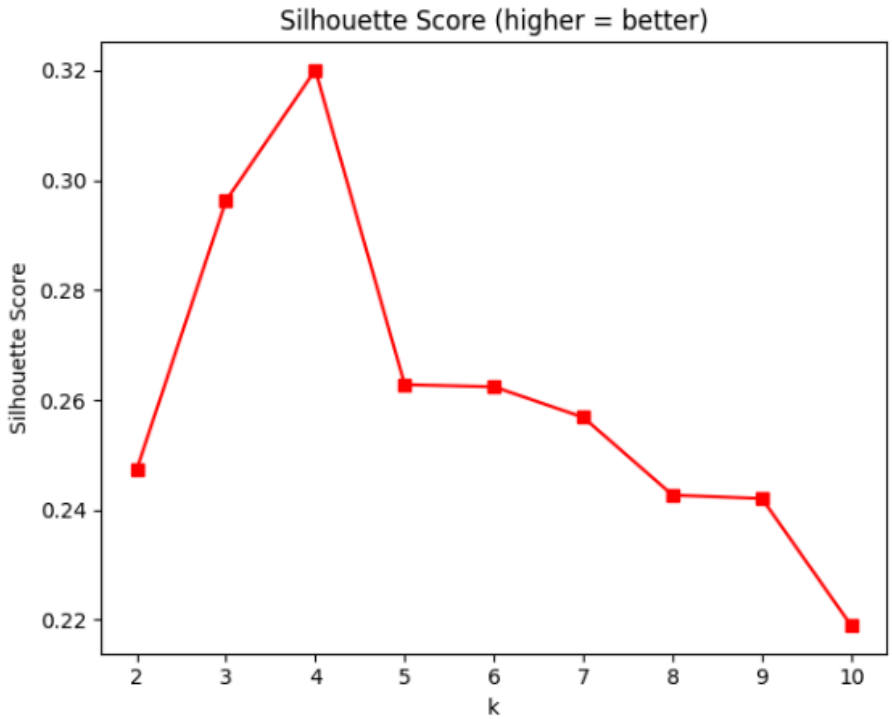



Figure 6. Hierarchical Clustering Dendrogram

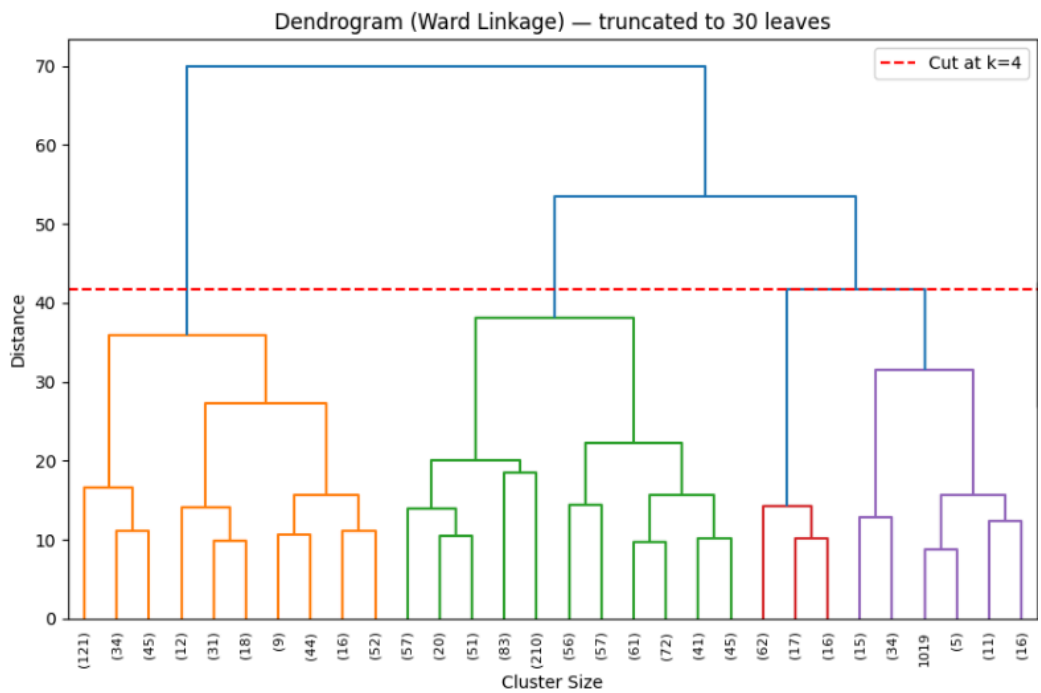



Figure 7. Clustering Comparison Chart

K-Means vs Hierarchical Agreement
Adjusted Rand Index = 0.429

| | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| 0 | 31 | 37 | 2 | 56 |
| 1 | 87 | 578 | 0 | 11 |
| 2 | 3 | 87 | 93 | 15 |
| 3 | 261 | 51 | 0 | 0 |

Figure 8. Cluster z-Score chart

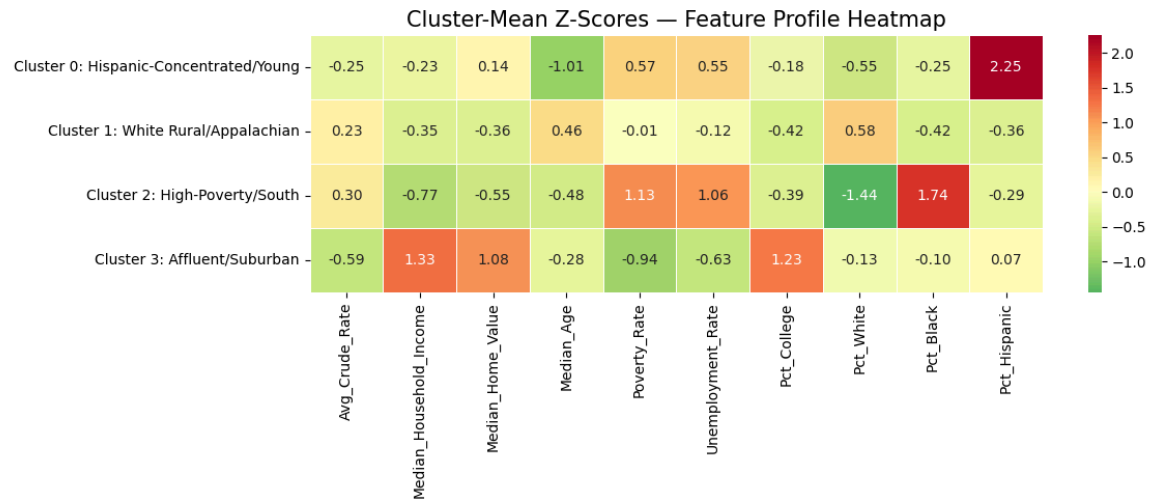



Figure 9. Elbow graph for DBSCAN eps value at 95th percentile

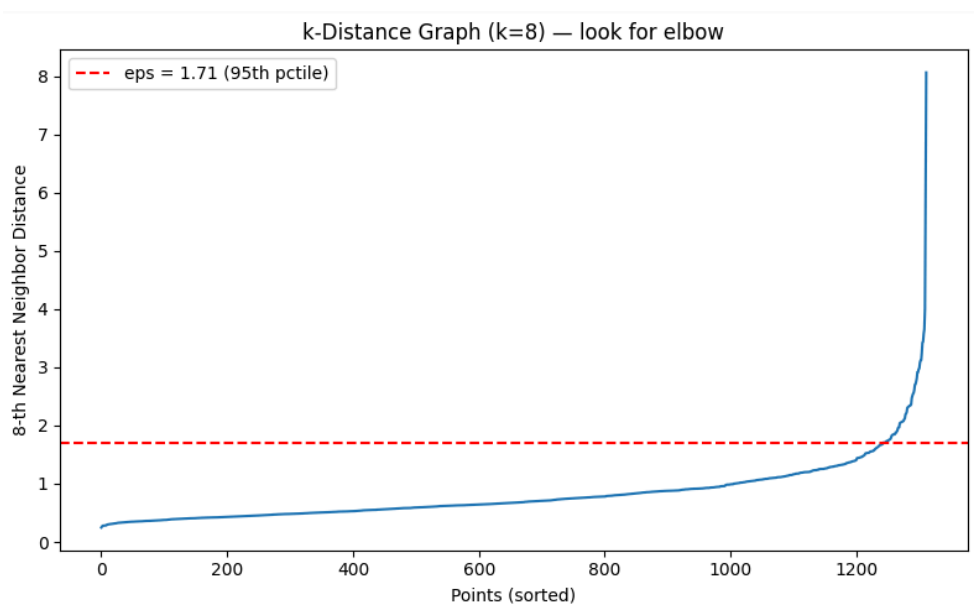


Figure 10. DBSCAN cluster and outliers

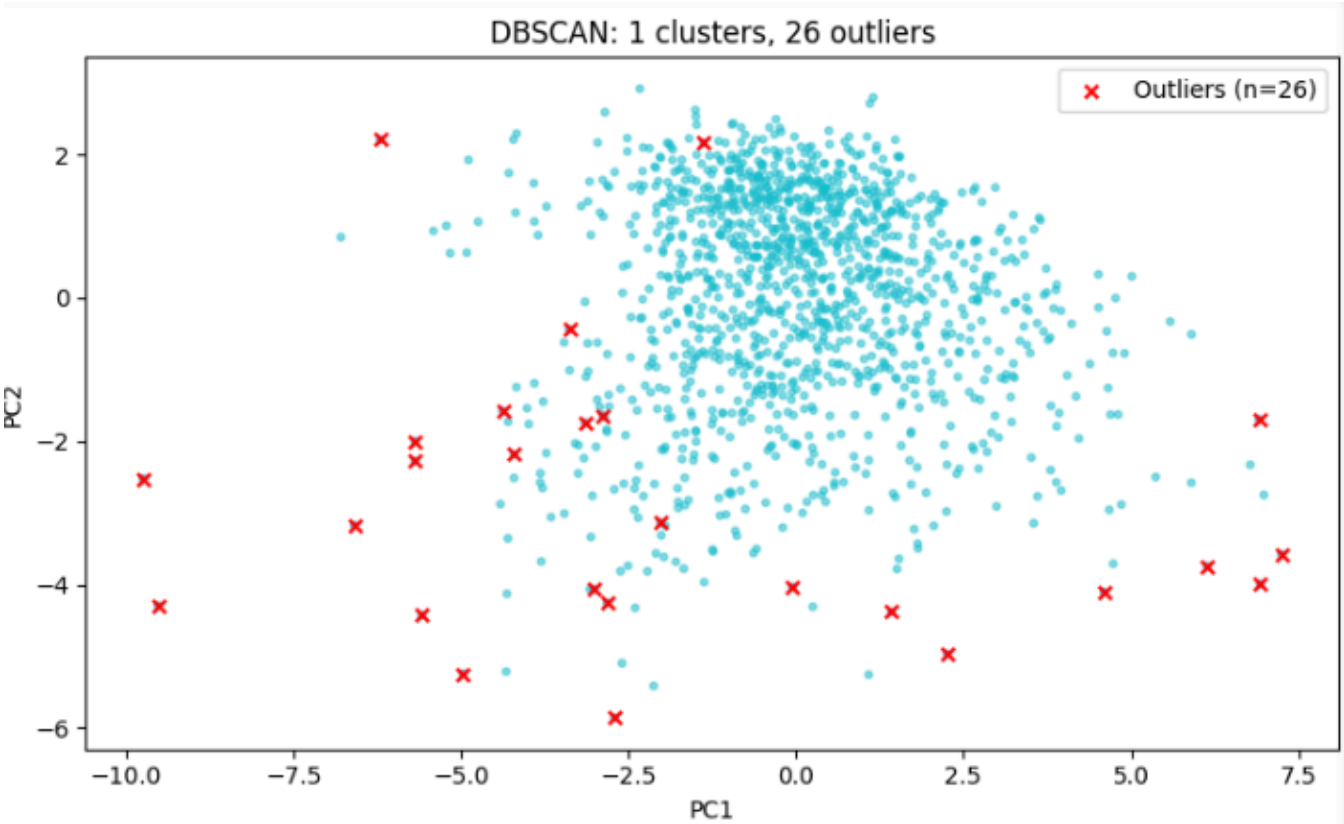



Figure 11. Sensitivity analysis ablation study for clustering

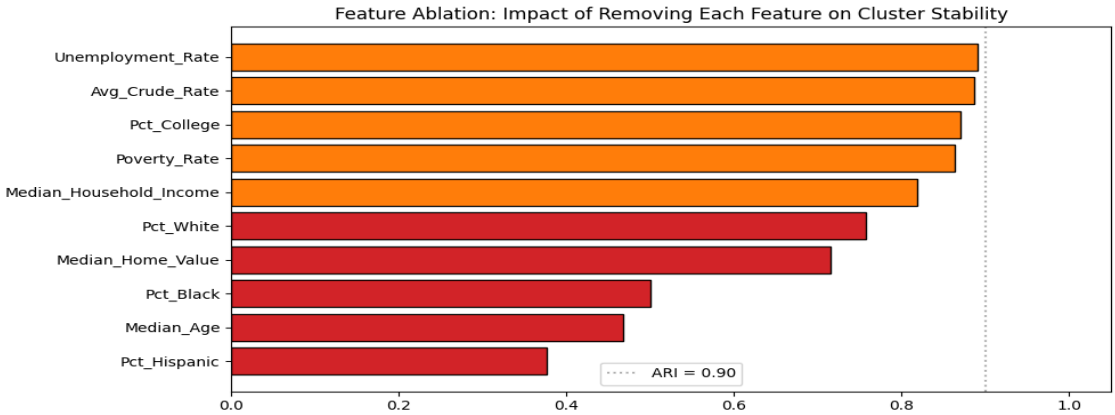


Figure 12. Geographic map of drug overdose mortality clusters and demographic data with DBSCAN outliers

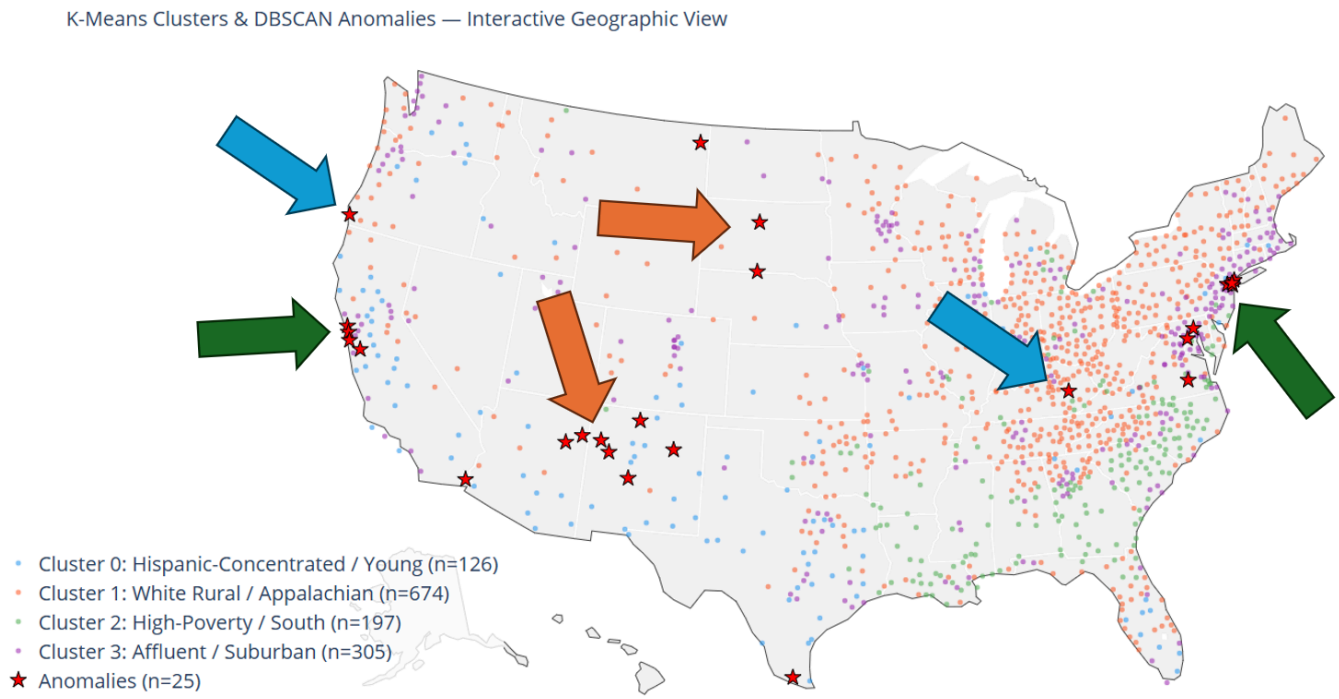



Figure 13. Dataset review

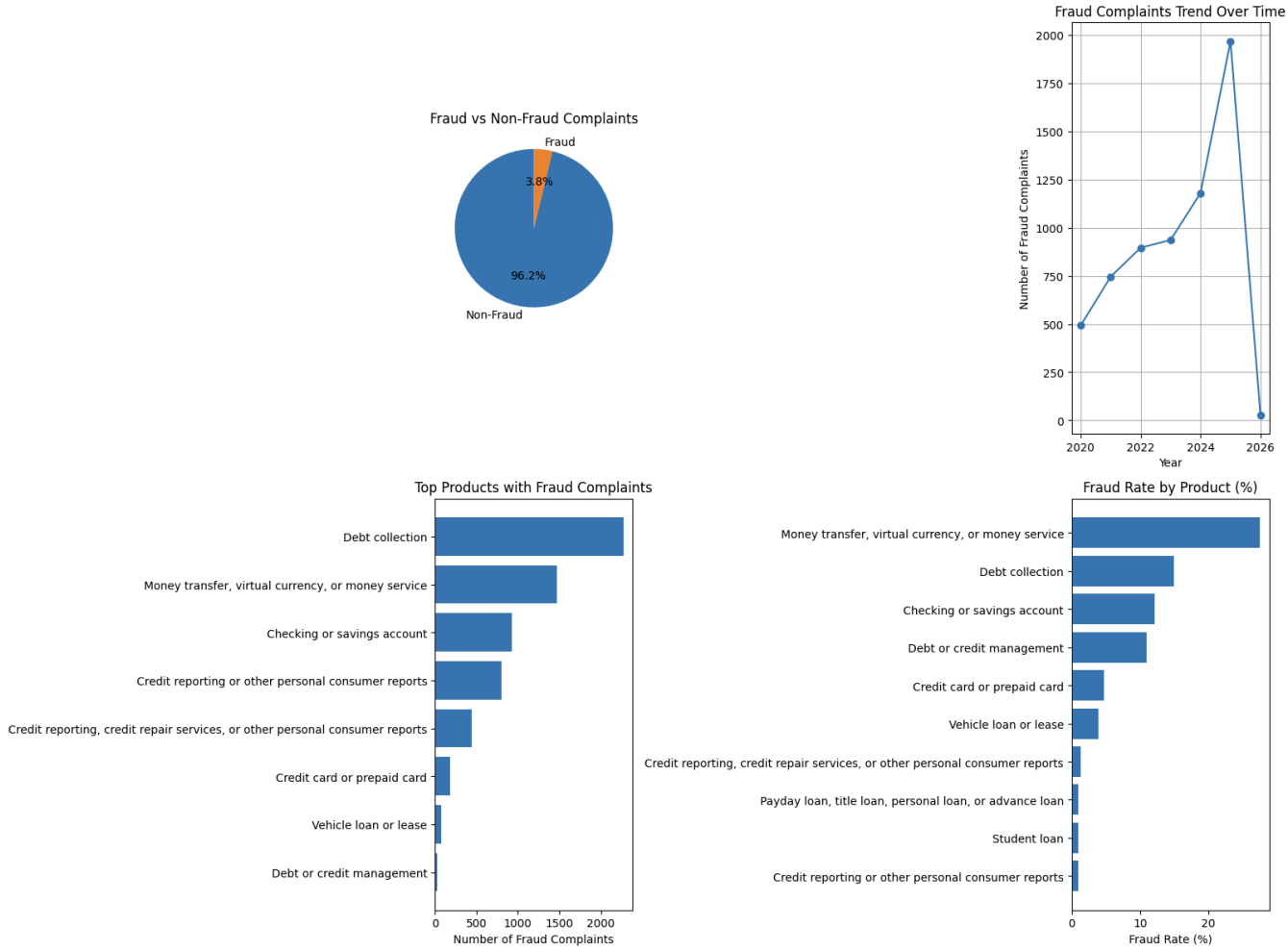


Figure 14. Logistic Regression Visual

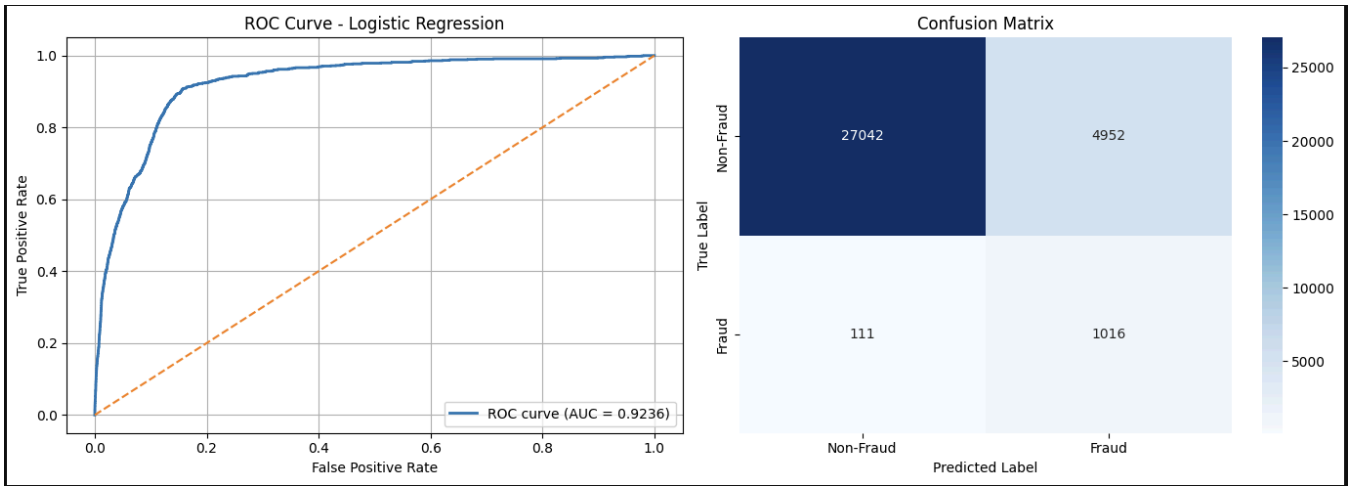



Figure 15. Random forest result Visual

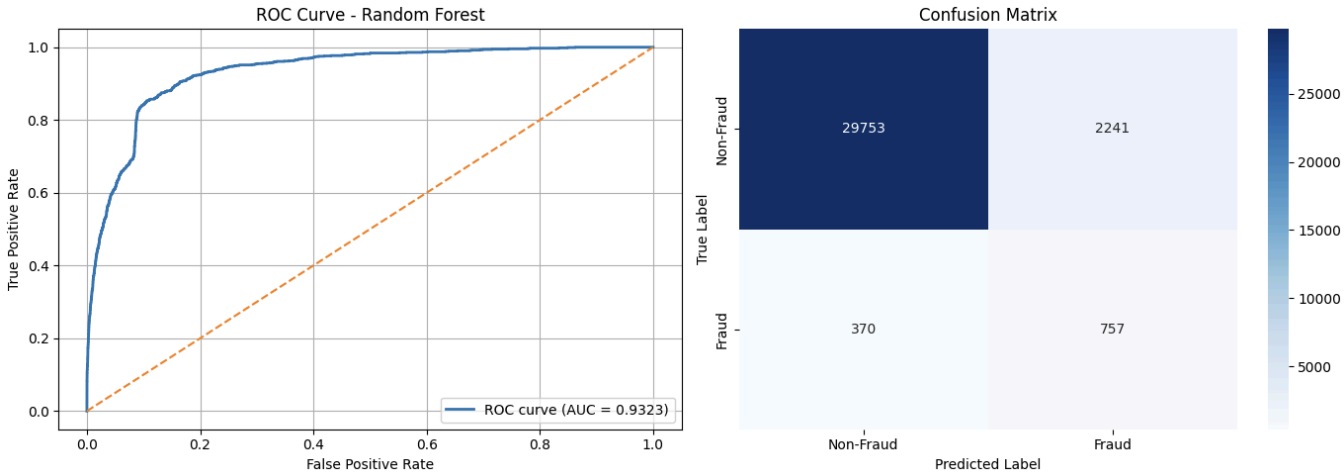



Figure 16. XGBoost result Visualization

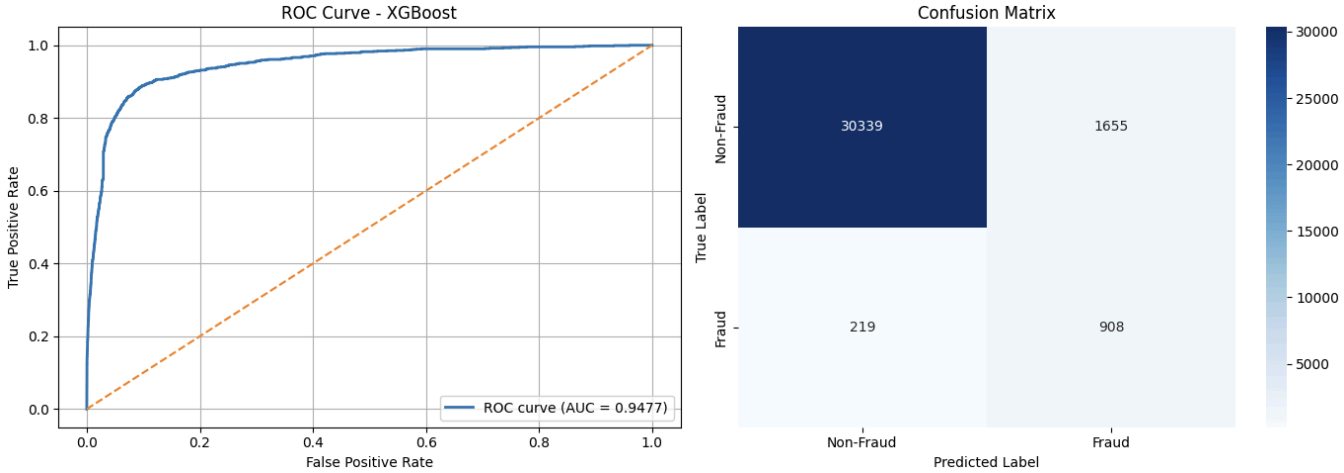


Figure 17. Model Results Compared

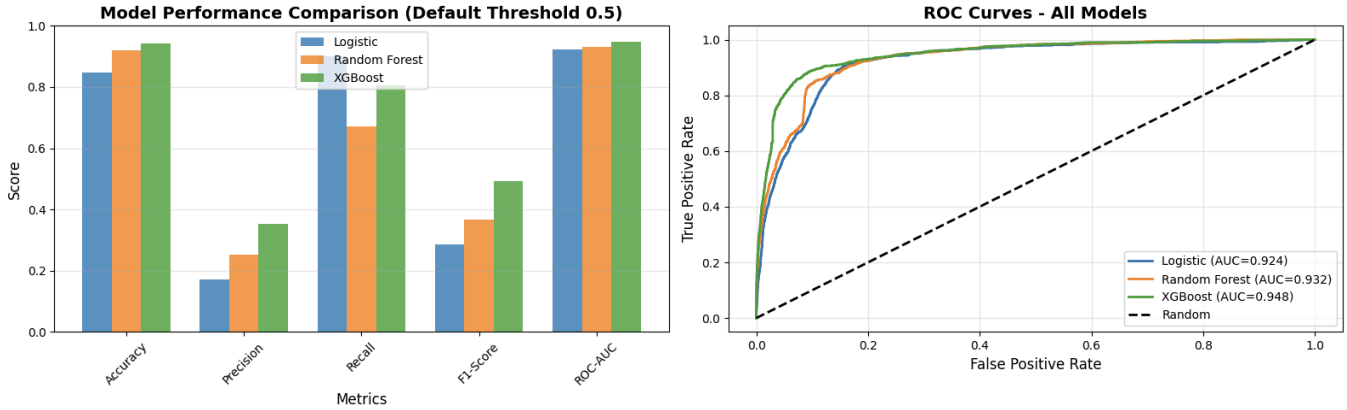



Figure 18. Ablation Study

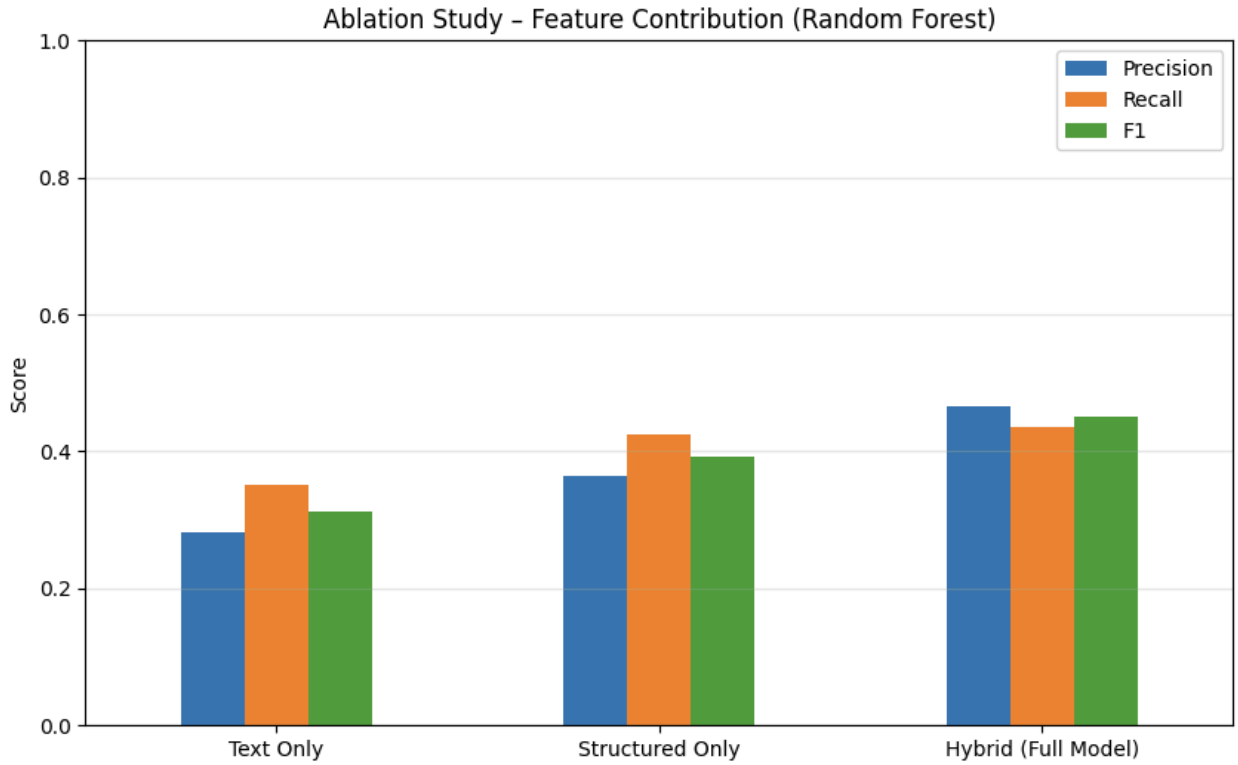



Figure 19. Hyperparameter Sensitivity

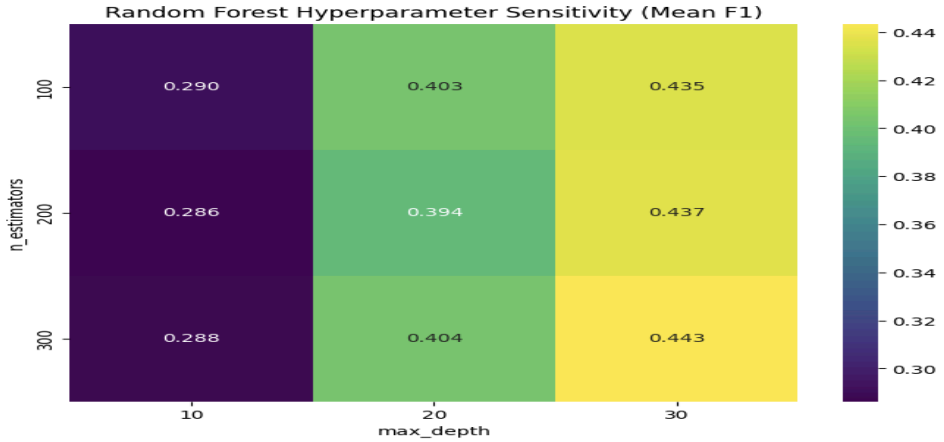


Figure 20. Smote Result

| Model | Threshold | Fraud Precision | Fraud Recall | Fraud F1 | AUC |
|---|---|---|---|---|---|
| RF (Original) | 0.6498 | 0.4659 | 0.4366 | 0.4508 | 0.9323 |
| RF + SMOTE | 0.4741 | 0.4586 | 0.5306 | 0.4920 | 0.9459 |
| XGB (Original) | 0.7788 | 0.5076 | 0.5049 | 0.5062 | 0.9472 |
| XGB + SMOTE | 0.1091 | 0.4252 | 0.7542 | 0.5438 | 0.9503 |

Key Findings¶

Applying SMOTE to the training data improved fraud detection performance for both models.

- Random Forest Fraud F1 increased from 0.45 to 0.49 (+9%).
- XGBoost Fraud F1 increased from 0.51 to 0.54 (+7%).
- XGBoost + SMOTE achieved the highest overall Fraud F1 (0.5438) and highest ROC-AUC (0.9503) of any configuration tested.

The primary effect of SMOTE was a substantial increase in recall. For XGBoost:

- Recall increased from 50.5% to 75.4%
- Precision decreased from 0.51 to 0.43
- The optimized threshold shifted from 0.78 to 0.11, reflecting the rebalanced probability distribution after oversampling.

This demonstrates the classic precision–recall tradeoff. SMOTE shifts the operating point toward higher fraud coverage at the cost of more false positives. In practical terms, XGBoost + SMOTE detects approximately 25% more fraud cases compared to the non-SMOTE configuration.

The improvement in ROC-AUC (0.9472 → 0.9503) is modest, indicating that SMOTE slightly enhances ranking ability. However, its primary benefit lies in enabling a higher-recall operating threshold.

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# Conclusion

SMOTE is a viable and effective strategy for this imbalanced dataset.

From a purely performance-driven standpoint, XGBoost + SMOTE is the strongest overall model, achieving the highest Fraud F1 and highest ROC-AUC across all experiments.

However, the final deployment decision depends on operational priorities:

- If the goal is balanced precision and recall, XGBoost without SMOTE (threshold = 0.78) provides a stable middle ground.

- If the priority is maximizing fraud detection coverage, particularly in a regulatory or screening context where missed fraud is costly, XGBoost + SMOTE is the preferred configuration.

Given the fraud detection objective and imbalance level (~3.8%), XGBoost + SMOTE represents the most effective model configuration overall.

Figure 21. List of features for the supervised learning from the CFPB

| **Column Name** | **Description** |
|---|---|
| narrative_clean | TfidfVectorizer narrative data |
| Product | One hot encoding of the product |
| State | One hot encoding State |
| Year | Numeric for Year |
| Month | Numeric for Month |
| Quarter | Numeric for Quarter |
| DayOfWeek | Numeric for Day of Week |
| narrative_length | Numeric count any element in the cleaned narrative |

Figure 22. List of specific reports from the Census ACS database

| | |
|---|---|
| B19013_001E: Median_Household_Income | B15003_024E: Professional_Degree |
| B17001_001E: Poverty_Universe | B15003_025E: Doctorate_Degree |
| B17001_002E: Population_Below_Poverty | B01003_001E: Total_Population |
| B23025_003E: Civilian_Labor_Force | B02001_002E: White_Alone |
| B23025_005E: Unemployed | B02001_003E: Black_Alone |
| B15003_001E: Education_Universe | B03003_003E: Hispanic_Latino |
| B15003_022E: Bachelors_Degree | B25077_001E: Median_Home_Value |
| B15003_023E: Masters_Degree | B01002_001E: Median_Age |

Figure 23. List of features in the Unsupervised learning dataset:

| **Column Name** | **Description** |
|---|---|

| | |
|---|---|
| Fips | FIPS Code, which is a unique identifier for geographic regions |
| State | The name of the State in the United States of America |
| County | The County name |
| Avg_Deaths | Average deaths across the years of data |
| Avg_Population | Average population across the years of data |
| Avg_Crude_Rate | Average crude death rate by overdose |
| Num_Years | Number of Years data is available for |
| Median_Household_Income | Median Household Income in the county |
| Median_Home_Value | Median home value in the county |
| Median_Age | Median age in the county |
| Poverty_Rate | Poverty rate in the county |
| Unemployment_Rate | Unemployment rate in the county |
| Pct_College | Percent of population with any level of college |
| Pct_White | Percent White |
| Pct_Black | Percent Black |
| Pct_Hispanic | Percent Hispanic |
| Total_Population | Count of population |
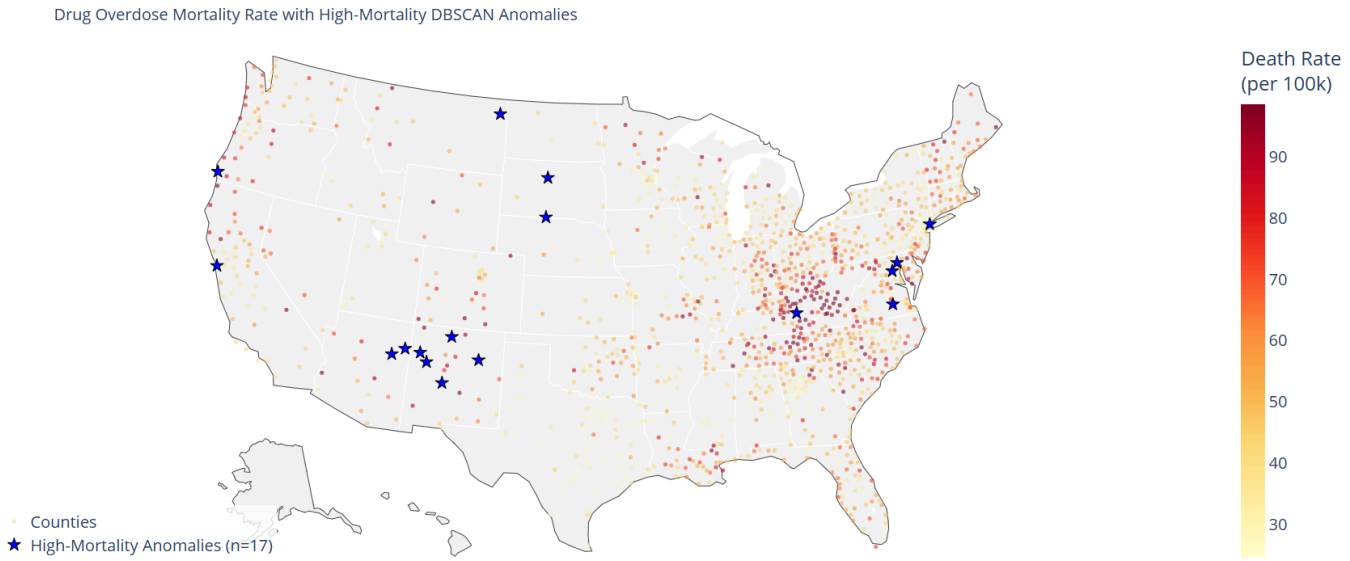
Figure 24. Geographic map of drug overdose mortality rate and DBSCAN outliers